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COURSE: SOFTWARE ENGINEERING

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ASSIGNMENT 02

Question no 01:

Describe the following in Software Engineering with real life examples

Concept Name: Generalization

- **Definition:**

Generalization is the process of identifying common characteristics among different entities and grouping them into a broader, more general category. In software, this helps reduce duplication by allowing shared features to be defined once in a "parent" class and inherited by "child" classes.

- **Real-Life Example:**

Think of the category "Vehicle." A car, a truck, and a motorcycle are all types of vehicles. Instead of describing each one separately from scratch, we generalize them under the "Vehicle" class, which includes common attributes like speed, fuel type, and capacity. Then, each specific type (Car, Truck, Motorcycle) can inherit these shared features and add their own unique ones—like "has cargo bed" for a truck or "has two wheels" for a motorcycle.

Concept Name: Object Class Aggregation Model

- **Definition:**

Aggregation models describe how complex objects are composed of simpler ones. It shows a "whole-part" relationship, where one class (the whole) contains other classes (the parts), but the parts can still exist independently.

- **Real-Life Example:**

Consider a "Library" system. A Library contains Books, Computers, and Staff. Each of these—Book, Computer, Staff—can exist on their own, but they are also part of the Library. If the Library closes, the books and computers don't cease to exist—they can be moved elsewhere. This illustrates aggregation: the Library is the whole, and the Books, Computers, and Staff are its parts.

Concept Name: Data Driven Modeling

- **Definition:**

Data-driven modeling focuses on how data flows through a system. It shows how input

data is processed step-by-step to produce output, often used in systems where data processing is the main activity.

- **Real-Life Example:**

Imagine an online tax filing system. A user enters their income and expenses (input data). The system processes this data to calculate deductions, taxable income, and finally the tax amount (output). The entire process is driven by the data the user provides, and the system's behavior depends on that data.

Concept Name: Event Driven Modeling

- **Definition:**

Event-driven modeling focuses on how a system reacts to external or internal events. It assumes the system is idle until an event occurs, which then triggers a specific response or transition.

- **Real-Life Example:**

Think of a doorbell system. It does nothing until someone presses the button (event). Once pressed, the system rings a bell (response). If no one presses the button, the system remains idle. This model is common in real-time systems like alarms, ATMs, or traffic lights.

Concept Name: State Machine

- **Definition:**

A state machine represents the different states a system can be in and how it transitions from one state to another based on events. It's useful for modeling systems that have a finite number of conditions or modes.

- **Real-Life Example:**

Consider a microwave oven. It has states like "Idle," "Cooking," "Paused," and "Finished." When you press "Start," it moves from Idle to Cooking. If you open the door, it transitions to Paused. When the timer ends, it goes to Finished. Each button press or event causes a change in state, and the system behaves differently depending on its current state.
